CSE422: Artificial intelligence

Assignment 04 [Spring 2026] v1.1

I highly recommended solving all these problems for a good preparation

Name:		**ID:**		**Section:**	

Probability

1. We construct a joint distribution table from three random variables: Device = {Laptop, Mobile}, User = {Student, Teacher}, Performance = {High, Low}

Device	**User**	**Performance**	**Probability**
Laptop	Student	High	0.12
Laptop	Student	Low	0.08
Laptop	Teacher	High	0.15
Laptop	Teacher	Low	0.05
Mobile	Student	High	0.2
Mobile	Student	Low	0.07
Mobile	Teacher	High	0.18
Mobile	Teacher	Low	0.15

(a) Calculate the marginal probability of a user being a teacher. [Basically, calculate P(teacher).]
(b) If you are given a mobile device, calculate the probability of it having a low performance. [Calculate P(Low | Mobile)]
(c) Are **User** and **Performance** conditionally independent of each other, given **Device = Mobile**?

2. A company uses an email spam filter. The filter correctly detects spam emails 90% of the time. It incorrectly marks a legitimate email as spam 4% of the time. In the company's inbox, 15% of all emails are spam. If the filter marks an email as spam, what is the probability that it is actually not spam?

3. A bank classifies loan applicants into two **risk categories** based on historical data. Among all applicants, 60% are typically **Low Risk** while 40% are **High Risk.** For **Low Risk** applicants, the bank has observed that 90% maintain **good credit history** and 85% have **stable employment**. Meanwhile, **High Risk** applicants show weaker financial profiles, with only 60% having **good credit history** and 50% maintaining **stable employment**.
The bank's model assumes credit history and employment status are **conditionally independent** given the risk category.

(a) What is the probability that a randomly selected loan applicant is a Low Risk applicant and has a good credit history?

(b) Calculate the probability that a randomly selected applicant has both good credit history and stable employment.
(c) An applicant has poor credit history and unstable employment. Based on the given probabilities, determine the most likely risk category for this applicant.

Naive-Bayes Classifier

4. A national park uses sensors and ranger observations to classify whether an animal sighting belongs to a rare species.

Sighting ID	Time of Day	Location	Sound Detected?	Rare Species?
A01	Morning	Lakeside	Yes	Yes
A02	Night	Deep forrest	No	No
A03	Evening	Deep forrest	Yes	Yes
A04	Morning	Grassland	No	No
A05	Night	Lakeside	Yes	Yes
A06	Evening	Lakeside	No	No
A07	Morning	Deep forrest	Yes	Yes
A08	Night	Grassland	Yes	No

Using the Naive Bayes classifier, predict whether the animal you saw last **evening, at lakeside, and heard barking** belongs to a rare species. [Hint: Basically, find out *P(Rare = Yes | TimeOfDay = Evening, Location = Lakeside, SoundDetected = Yes)* and *P(Rare = No | TimeOfDay = Evening, Location = Lakeside, SoundDetected = Yes).* Predict the one with higher value.]

5. Consider the following dataset, where x_1 and x_2 given y belong to a multinomial distribution:

x_1	x_2	y
0	1	0
1	0	0
1	1	1
0	1	0
0	0	1
1	1	2

(a) Using naïve bayes, calculate the most likely outcome of y given $\mathbf{x_1} = 1$ and $\mathbf{x_2} = 1$
(b) Consider that a new feature x3 with continuous values, that is x3 $\in \mathbb{R}$, is to be added to the dataset. How can the naive Bayes classifier be applied with x3.

Linear Regression and Logistic Regression

6. You are given the following dataset. You want to fit a linear regression model on this dataset with initial parameters m = 2 and b = 1.

X	Y
1	5
3	12
5	15

(a) Calculate the **Mean Squared Error (MSE)** for this model on the dataset.
(b) Perform TWO iterations of **gradient descent** to update both m and b, assuming the learning rate = 0.05.
(c) For these specific values, **explain** which loss function would be more suitable (i) $(y_i - y')^3$ **or** (ii) $|(y_i - y')|$. Use initial parameters.

7. Consider the following regression dataset:

Sample	Hours studied (x1)	Hours slept (x2)	Test score (y)
1	1	3	5
2	2	2	4
3	0	4	1
4	3	1	9

You decide to use linear regression to solve this problem. For the weights, choose any value between [0.1, 0.9]. The learning rate must be between [0.01, 0.09].

(a) Calculate the **Mean Squared Error (MSE)** for this model on the dataset.
(b) Perform TWO iterations of **gradient descent** to update the parameters. Report the updated values.
(c) Using the updated values, find out the predicted test score for a student who studied 2.5 hours and slept 2 hours.

8. You want to fit a logistic regression model for the following classification dataset. You start with weight = 3 and bias = 1.

X	Y
0.2	0
0.4	0
0.8	1

(a) What type of classification is this? What would be a suitable loss function? [Hint: BCE]
(b) Calculate the amount of error using your chosen loss function.
(b) Perform two iterations of **gradient descent** to update the model parameters, assuming the learning rate = 0.1.

9. Consider the following classification dataset:

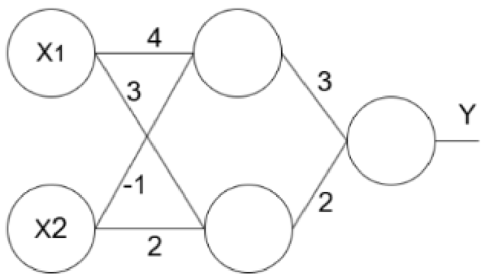
Sample	Cholesterol level (x1)	Exercise hour (x2)	Has Disease? (y)
1	6.5	3	0
2	6.2	0	1
3	5.0	4	0
4	7.8	1	1

You decide to use logistic regression to solve this problem. For the weights, choose any value between [0.1, 0.9]. The learning rate must be between [0.01, 0.09].

(d) Calculate the error using **Binary Cross-Entropy (BCE)** for this model on the dataset.
(e) Perform TWO iterations of **gradient descent** to update the parameters. Report the updated values.
(f) Using the updated values, predict whether a person with a cholesterol level of 4.0 and 1 hour of exercise will have disease.

MLP (Neural Network)

10. Consider the following feed-forward neural network. Here the circles represent a perceptron (neuron), which is a weighted sum followed by an activation function. The hidden layer neurons use ReLU activation while the output layer uses sigmoid activation.



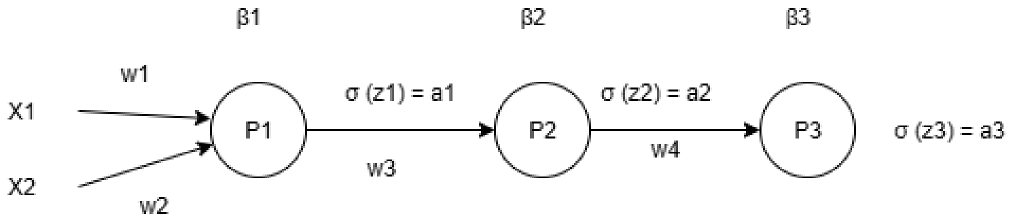
If x1 = 5 and x2 = 1, apply forward propagation to calculate the value of y.

11. Suppose we are training a Feed-forward Neural Network with ONE feature. Network structure is as follows:
Input feature, x1 = 1
Hidden layer: 2 Neurons, weights are - 0.3 and 0.4, ReLU activation
Output layer: 1 neuron, weights are 1 and 2, Sigmoid activation
Loss function: Binary Cross Entropy

(a) Draw the neural network.
(b) Run the forward propagation and find the output.

(c) If given output y = 1, calculate the error using the loss function.
(d) Update all weights in the neural network using back propagation. You must show the derivatives. Take learning rate = 0.1.

12.



Suppose you are given the above architecture as an MLP for some inferential task. Your test set looks like x1 = 1, x2 = 2, and y = 1 (true label). Furthermore, the network assumes the weights and biases are initialized as w1 = 0.1, w2 = 0.5, w3 = 0.5, w4 = 0.2, β1 = 0.5, β2 = 0.5, and β3 = 0.5, respectively as part of its initial hypothesis. Moreover, P1, P2, and P3 are the three perceptrons whose non-linear outputs are a1, a2, and y', respectively. The Loss function is Binary Cross Entropy (BCE).

(a) Write the equation of the loss function.
(b) Perform one pass of forward propagation and determine the predicted output for the given test set.
(c) For one pass of backward propagation, first derive the expression for $\partial L/\partial w4$. Then use the chain rule to derive the expression for $\partial L/\partial w1$. Finally, calculate the quantitative values.

Thank you for your hard work throughout the semester.

Newton's third law – the only way humans have ever figured out of getting somewhere is to leave something behind.